MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) THIRD SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

MG3DSECSC202 – Introduction to Embedded Systems

Duration: 2 Hrs Maximum Marks: 70

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions
Answer All Questions
Each question carries 2 marks

1. Classify memories into volatile and non-volatile. [U][CO1]
2. List two everyday examples of embedded systems. [U][CO2]
3. Identify the difference between MCU and CPU. [U][CO3]
4. Define embedded system in context of mobile phones.[U][CO4]
5. List two comfort-related embedded applications in cars. [U][CO4]

[5x2 = 10]

Part B

Short Answer Questions
Answer any 6 Questions
Each question carries 5 marks

6. Explain why hierarchy is essential in modern computing systems. [U][CO1]
7. Describe key features of embedded systems with examples.[U][CO2]
8. Analyse stack pointer operation in embedded memory organization. [An][CO3]
9. Analyse the importance of low-power protocols like ZigBee in WSN. [An][CO4]
10. Summarize steps of program execution when an interrupt occurs. [U][CO1]
11. Describe the importance of feedback in ES model. [U][CO2]
12. Analyse the effect of resolution on accuracy of ADC. [An][CO3]
13. Describe the importance of embedded safety systems in robotics. [U][CO4]

[6x5 = 30]

Part C

Very Short Answer Questions
Answer any two Questions
Each question carries 15 marks

14. Explain interfacing methods of sensors (temperature, light, proximity) with MCU including circuits and applications. [An][CO3]
15. Discuss Brain–Machine Interfaces (BMI) with case studies of assistive technologies. [A][CO4]
16. Discuss MCU word sizes (8/16/32 bit) and their impact on performance and application domain. [U][CO2]

[2x15 = 30]